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Are You In Suspense?

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resources will now need to share a single hardware platform. In such a situation, capabilities provided by the foundation platform should be scalable. It should be able to accommodate legacy applications and also allow development of new applications in conjunction with legacy applications," explains Thilak Kumar, regional engineering manager, Wind River.

Tackling urban positioning. Getting your GPS device to work in an urban environment is difficult if you are using a GPS module without a network connection. The biggest challenges with urban positioning are lack of a direct line-of-sight due to tall buildings

GPS spoofing

GPS spoofing is one way to dupe a GPS receiver and literally throw it off course. This is done by emitting a counterfeit version of the signal emitted by a GPS satellite at a slightly higher power. This causes the GPS receivers in the area to pick up the fraudulent signal, which then goes on to make the module show a faulty position. It is very dangerous if it messes up the navigation systems in a ship, plane or autonomous vehicle. Point to note, this is different from GPS jamming, which is pretty boring compared to what spoofing does.

around you, and going indoors with cuts in access to all satellites.

The tall buildings make it very difficult for the GPS receiver to get data from all the GPS satellites in the area, which results in a longer TTFF. If you are indoors, your receiver will be unable to hear from a satellite unless you are near a window – not really the most favourable experience. How to tackle this?

A-GPS is one of the best things that have ever happened to positioning since GPS itself. It allows your device to gather data from the WWRN servers for four hours in advance. Once you get this data on your phone, your device no longer needs to connect directly with the satellites for the ephemeris data. Now you might think that since there is only four hours worth of data, you might need to go out every four hours to get fresh air and get the latest in ephemeris data. Not needed. This is because A-GPS sends data from the WWRN servers through any wireless connection of your device (such as telecom and Wi-Fi).

How are components developing for the future?

Newer components from vendors use specialised low-power analogue-to-digital converters and LNAs, and electronics circuit PLLs. The designs are also such that power supply and clocks are controlled for each separate block function. Some of the new chips also do not require individual sensor drivers as their CPUs have a sensor control function to take care of it.

Advanced low-power technology means never having to maintain full power to achieve maximum performance or turning the GNSS receiver completely off to save power.

"It breaks the old GNSS performance vs power trade-off through the fusion of a new high-performance GNSS engine, advanced power management and a smart sensor interface to achieve high-sensitivity hot-start conditions for fast location fixes," adds Reuben.

Regarding the multiple choices of satellite constellations that are available up there, "We recommend that designers use multi-constellation GNSS receivers that support as many constellations as possible. Multi-constellation receivers support concurrent operation and can either acquire or track using a mixture of satellites from any of the constellations supported. As new constellations become available, system designers do not have to choose from selecting one constellation or another. The best solution will support concurrent use of all constellations supported," explains Walter Rivera, Manager, Product Marketing and John Hutson, Sr Manager, IC Design Engineering, Broadcom. ●

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